

we have uncovered is amazing to me, as I would have expected predictive patterns in markets to change more over the longer term.

Is the implication then that models don't get dropped even if they perform poorly?

It takes a tremendous amount of deterioration to drop a model. We don't react to the short-term results of a model because the current year performance of any single model is simply not at all predictive of the next year's performance. What is predictive is how the model performed over the entire 31 years. The extra 3 percent of data provided by the most recent year doesn't make much difference in how a model has performed over the entire training period.

Your assets under management in the futures program alone have been as high as \$5 billion. Is capacity a problem? Do you have to make changes to accommodate larger assets under management?

One of the changes we have made from the early years, which has greatly expanded capacity, is to execute trades throughout the trading session as opposed to only on the opening. Another change that has increased our capacity is that we have shifted the allocation process to give greater weight to more liquid markets. We trade a larger percentage in stock indexes and interest rates than we used to and a smaller percentage in nonfinancial futures contracts. Although this change has reduced our diversification, we were quite willing to make it because there is a strong pattern for our edge to be greater in more liquid markets. So besides increasing capacity, the shift to allocating a greater percentage to more liquid markets has also improved performance.

Are you then sizing your positions in each market based on relative liquidity?

We started shifting our weightings to the more liquid markets in 2006 and about six months ago we switched our risk weightings to be exclusively based on liquidity, with the one exception being the S&P, which has very substantial excess capacity.

The other important point that needs to be made about capacity is that it is not static; it moves around a lot with changes in volume and volatility